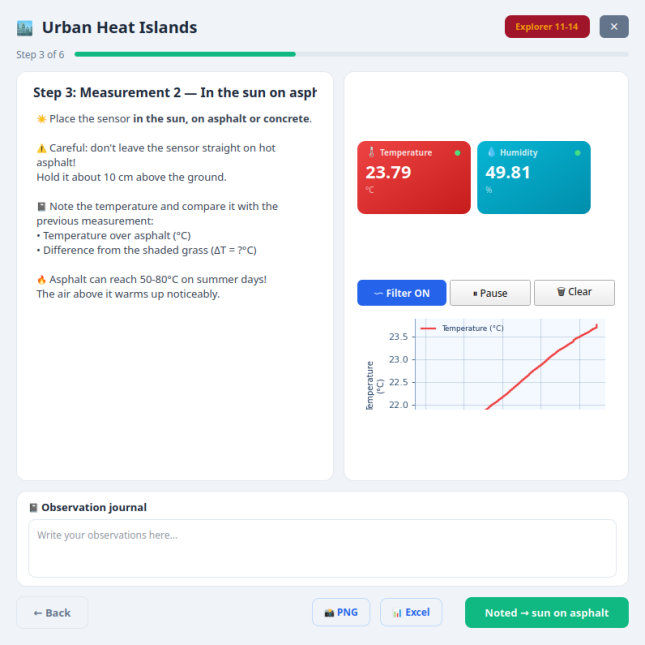
Urban Heat Islands

A measured walk around the school grounds, and the heat map it produces

Level: Explorer — ages 11–14 (Romanian school years V–VIII) • **Time in class:** 25 min, two or three 50-minute lessons • **Sensors:** BME280 (temperature, humidity, pressure)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer — ages 11–14 (Romanian school years V–VIII). The short route also works for year IV, without the drift correction. The most direct curriculum entry points are **year V** (geography: the Earth's atmosphere) and **year VIII** (physics: heat transfer).

Assumed knowledge

Reading a number with one decimal place, computing a difference and the arithmetic mean. Orientation on a plan of the schoolyard. For extension, in addition: proportional distribution of a correction. For support: only "warmer / colder" and "in the sun / in the shade".

Time in class

Short: 25 min. Full: two 50-minute lessons (1 h 40 min). Project: three 50-minute lessons (2 h 30 min).

Teacher preparation

30 min — **walking your own route, box and watch in hand**, choosing the six spots, printing the schoolyard plan, cutting the white card screens, notifying school staff, checking the forecast.

Sensor settling time

A full two minutes at each spot, plus **five minutes** at the first exit from the building. This is the single most common reason for poor data.

Sensors and boxes

BME280. **A single box** per group: the activity is built for one box that walks the route. A second box, left fixed in the shade as a *reference station*, is strongly recommended; before setting off, the two boxes sit side by side for five minutes so you can compare them. For *simultaneous* measurement, follow up with the *Heat Islands with Several Boxes* app.

Three routes

Short	25 min: the written prediction, the protocol displayed, the calibration, grass in shade, asphalt in sun, indoors, the return to the grass, the two differences. Exactly the eleven steps in the app, with three locations under a minute's walk apart.
Full	Two 50-minute lessons: lesson 1 is designing the route on the printed plan and negotiating the protocol, without the sensor; lesson 2 is the walk itself, with the loop closed and the export.
Project	Three 50-minute lessons: the two above, plus the analysis lesson — the drift correction, the class heat map and the reasoned proposal for the schoolyard.

1. Where this fits in the curriculum

Students walk their own school's grounds on foot, measure the air temperature above grass, above asphalt, in sun, in shade and indoors, and discover that a single place shelters several microclimates that differ by several degrees from one another. The phenomenon is real, published and actively researched, and their method is a simplified variant of the method used by urban climatologists. **Let us be precise about the name:** what students measure in the schoolyard are *microclimates*, differences from spot to spot driven by surface and shade. **The urban heat island**, in the climatologists' sense, is the difference in *air* temperature between a town and the countryside around it, and it is largest *at night*. The schoolyard is its miniature model, with the same mechanisms. The activity fits:

- **Geography** (*Geografie*, most direct at year V) — at **year V**, the Earth's atmosphere: air temperature, the factors that change it and the idea of weather. This is the most direct curriculum entry. At **year VIII**, the geography of Romania covers human settlements and the city, and the result measured in the schoolyard applies there — but that is an *application* of the measurement, not a curriculum entry about the urban microclimate. The fieldwork technique itself — route, protocol, representation on a map — is a geography skill at every year of lower secondary.
- **Physics** (*Fizică*, year VIII) — at **year VIII**, thermal phenomena cover heat and **heat transfer by conduction, convection and radiation**: dark asphalt absorbs solar radiation, then warms the air above it by conduction and convection. Here the activity sits exactly on the syllabus, together with the idea that a body exposed to the sun heats up until its losses equal its gains. At **year VI** it links to temperature and thermal equilibrium.
- **Biology** (*Biologie*) — plant transpiration and the role of the leaf. That watered grass cools the air through evaporation is an *application* of this subject, not a separate curriculum entry; but it is the most convincing application a twelve-year-old can see, because they measure it themselves.
- **Mathematics** (*Matematică*, years V–VII) — the difference of two decimals and the arithmetic mean at **years V–VI**; plotting points and reading a plan to scale at **year VII**; proportional distribution of a correction, on the extension route, is exactly direct proportionality from the syllabus.
- **Cross-curricular** — the distinction between a *surface* and the *air above it*, the habit of noting the time alongside the number, the care not to contaminate your own measurement by standing next to the sensor, and formulating a proposal for a real place, backed by your own data.

The core skill is **controlled comparison in an uncontrollable environment**. Outdoors, nothing can be held constant, so students must standardise *their own* procedure — the same height, the same shading of the sensor, the same waiting time — and understand why. It is a finer idea than the controlled laboratory experiment, and it is exactly what fieldwork teaches. And wherever there is a choice, change **one thing at a time**: grass in sun against grass in shade says what shade does; asphalt in sun against grass in shade changes *two* things at once and cannot say which

of them mattered.

This activity prepares the multi-box app

The two heat-island resources are a pair, and they are done in this order. Here, with **a single box**, students learn the field protocol: the same height, the same shading, two minutes of waiting, an exclusion circle, the time noted — and, by closing the loop, they learn that every measurement has a scatter of its own, which decides what you are allowed to claim.
The *Heat Islands with Several Boxes* app takes the next step: **several boxes, at the same moment**, which removes exactly the problem the closed loop repairs here — that the places were not measured at the same time. There, the scatter is measured directly, by putting the boxes side by side five minutes before setting off — which you do here too, on a small scale, with the walking box and the reference station; it is called the *detection limit*, and the rule students learn is the same one as in the investigation box below.
Do not reverse the order. A class handed six boxes before holding one still for two minutes at 1.2 m produces six data sets that cannot be compared.

2. Time plan

The app's wizard has eleven steps and takes about twenty-five minutes, *if* the grass, the asphalt and the classroom are all under a minute's walk apart. What turns the activity into a genuine investigation is a planned walk of a route, with a return to the first spot and a class heat map, and that takes three lessons.

Route	Time	What happens
Short	**25 min**	The written prediction and the class vote (3), the protocol read out and displayed on the board (2), the calibration — with a single box, one captured value and on (3), grass in shade — the reference spot (4), the second prediction and asphalt in sun (4), indoors (3), grass in shade again, which closes the loop (3), the analysis table, the "How sure are we?" step and the closing sentence (3). Exactly the eleven steps in the app, nothing more.
Full, lesson 1	**50 min**	The written predictions and the vote (10), the mechanism — why asphalt heats up and why grass cools down (10), designing the route on the printed plan, six spots each with a written reason (15), negotiating the shared protocol and writing it on the board (10), a "cold" rehearsal of a single reading, right by the door (5). **No trip outdoors.**
Full, lesson 2	**50 min**	The safety briefing and getting equipped (5), the calibration — the two boxes side by side, in the shade, five minutes — and starting the reference station (8), walking the route — six spots, four minutes each, that is $6 \times 4 = 24$ minutes —, returning to spot 1 to close the loop (5), back to the classroom, the analysis table, the "How sure are we?" step, the app's conclusion and the export (8).
Project, lesson 3	50 min	The drift correction applied row by row (10), the class heat map (15), the two differences compared with the drift and the offset, the ranking of surfaces and the link to the town's heat island, which shows at night (13), the reasoned proposal for the schoolyard, with a number from your own data (12).

The full route means **two 50-minute lessons** — an hour and forty minutes — and the project route adds the third lesson, so **three 50-minute lessons**, two and a half hours of actual time. Time spent outdoors totals about **thirty-five minutes**: the calibration, the 24 minutes of measuring and the return to spot 1. That is exactly why the loop closes — in half an hour of a sunny morning, the whole schoolyard can warm up by more than the difference between two locations.

Choose the day and the time on purpose

The heat-island signal in a school playground is largest on a **clear, calm and sunny** day, and it is largest **in the early afternoon**, when the asphalt has had hours of sun. On a cloudy or windy day the differences can drop to a few tenths of a degree, and the class will conclude, reasonably but wrongly, that the phenomenon does not exist. Check the forecast a week ahead and be ready to move the lesson to another day.

If you cannot, do it anyway and turn the null result into the topic: *why does wind destroy the effect?* The answer — mixing of the air — makes an entire lesson on its own. Tell the class *beforehand*, though, that a small number is a possible outcome, so it becomes an observation rather than a disappointment.

3. Materials

What you need	Per group	Notes
MagicBox with BME280	1	The box that walks the route. Temperature and humidity are used; humidity is what makes evapotranspiration above the grass visible.
A second box, as a reference station	1 per class	Stays in one shaded, ventilated spot the whole time. It records how much the *whole* place warmed up while you walked, and that number must be subtracted before comparing locations. **Strongly recommended.**
Laptop with the app	1	Carried by one named person, who carries nothing else. Screens are hard to read in the sun: choose a shaded reading spot or turn up the brightness before you go out.
White screen for the sensor	1	A piece of white card, a folded sheet or a white envelope, held above the box. Without it, sun falling on the casing gives a reading several degrees too high. **The single most important item on this list.**
Tape measure or metre rule	1	To fix the measuring height at 1.2 m and to note the distance to a wall or a tree. Height is *measured*, not guessed.
Stand, stick or crate	1	So the sensor sits at the agreed height, not in someone's warm palm.
Schoolyard plan, printed	1	A4, one per group, with the spots numbered. A satellite-map screenshot works very well and makes the surfaces visible.
Clipboard, sheet and **pencil**	1	Pencil, not pen: wind and damp defeat pens.
Watch or phone	1	**Every reading carries a time.** Without times, the drift correction is impossible.
Chalk or numbered tokens	6–8	To mark the exact spot of each stop, so a second walk uses the same points.
Infrared thermometer	1 (optional)	Reads the *surface* temperature. Makes the contrast between the very hot asphalt and the much cooler air above it visible on the spot, and is the best possible improvement to the activity.
Reflective vests	1 / student	For any route that comes near a car park, a delivery ramp or the school gate.
Hats, sunscreen, water	1 / student	For a walk on a sunny, warm day — exactly the conditions that give the best data.

Preparation the day before: walk your own route, box and watch in hand, and time it. Choose six spots that really contrast — grass in shade, grass in sun, asphalt in sun, asphalt in shade, beside a sunlit wall, indoors — and check that every one can be reached safely and that none is near an air-conditioning vent, a kitchen extractor or a car with its engine running; all of these give spectacular readings that mean nothing. Print the schoolyard plan. Confirm you are allowed in

every part of the grounds and tell the office and site staff that a group will be outside. Charge the laptop, cut the white card screens and, at the end, check the forecast and prepare the **indoor route** from the fallback box: it works surprisingly well.

4. Scientific background for the teacher

The energy balance of a surface

Any surface — grass, asphalt, roofing — must get rid of the energy it receives. Per unit area,

$$Q^* = Q_H + Q_E + Q_G,$$

where Q^* is the net absorbed radiation, Q_H is the heat given to the air (this is what warms the thermometer), Q_E is the latent heat carried away by evaporating water, and Q_G is the heat conducted into the material and stored there. The net radiation itself is

$$Q^* = (1-\alpha)\,S{\downarrow} \;+\; L{\downarrow} \;-\; L{\uparrow},$$

with $S{\downarrow}$ the received solar radiation, α the **albedo** (the reflected fraction) and L the infrared exchange.
Dark asphalt has an albedo of about 0.05–0.15, so it **absorbs** 85–95% of the light that falls on it; fresh grass sits at 0.20–0.25, and light concrete at about 0.30. On top of that, a paved surface has no water to evaporate, so Q_E collapses towards zero and almost all the absorbed energy has to leave through Q_H and Q_G — that is, heating the air and charging the material. **The energy a lawn would have spent evaporating water is delivered to the air by asphalt instead.** That is the mechanism of the urban heat island in one sentence.

4.1 Why grass is cool: latent heat

Evaporating one kilogram of water absorbs about 2.45 MJ. A lawn that transpires a few millimetres of water a day is, therefore, a considerable air conditioner, one that uses no electricity and makes no noise. Because this energy leaves through the Q_E term, it does not show up as a rise in temperature.

Students can see the mechanism, not just hear about it, if they record **humidity alongside temperature**: above wet grass the air is measurably both more humid *and* cooler, and the two observations together make the explanation visible. Expect 5 to 15 percentage points more relative humidity above shaded grass than above sunny asphalt. In a drought, when the grass turns yellow and stops transpiring, the difference shrinks — which is itself a testable prediction.

4.2 What the sensor actually measures: the air, not the surface

This is the central trap of the activity and it is worth saying three times. The BME280 measures the temperature of the *air around it*, not the temperature of the surface it sits on. A sunlit asphalt surface can pass 50 °C, while the air 1.2 m above it reads 28 °C. Air is heated from below, by conduction and convection, and the effect weakens quickly with height; that is why the difference between two locations is always much smaller in air than at the surface.

The order of magnitude is this: in air, with the sensor correctly shaded, the difference between grass in shade and asphalt in sun is usually 1 to 3 °C, and that between sun and shade on the same surface 0.5 to 2 °C; at the *surface*, the same pair often differs by 25–30 °C, and that is exactly the figure shown by a satellite thermal image or an infrared thermometer. These are two different quantities. Confusing them is the single most common error in all public communication on the topic, and it is why students predict ten or twenty degrees for the air.

The practical consequence: **the measuring height must be the same at every location**,

otherwise the comparison stops meaning anything. If you have an infrared thermometer, take *both* numbers at the asphalt spot. The contrast between them is the most memorable moment of the whole activity.

4.3 Radiation error and the white screen

In shade and with moving air, casing temperature is a good approximation of air temperature. In direct sun it is not: the casing absorbs solar radiation and warms above the surrounding air, and errors of 3 to 8 °C are common. This is the **radiation error**, and it is bigger than every real difference you are looking for: an unshaded box in the sun against one in the shade "discovers" a five-degree heat island that is really just its own warmed casing. Weather services solve this with a white, ventilated screen, and the classroom substitute is the white card screen. With a screen, the real sun-versus-shade difference in the air drops to 0.5–2 °C.

Students should understand that **shading the instrument while it measures above a sunlit surface is not cheating**. On the contrary: without a screen they measure their own box, not the air, and the difference obtained between grass and asphalt would be an artefact. The rule applies *at every location, including the already-shaded ones*, so the treatment is identical; shading only where the sun beats down introduces exactly the difference you are trying to measure.

4.4 Thermal inertia, and why the time of day matters

Concrete and asphalt have a specific heat around 800–900 $\mathrm{J\,kg^{-1}K^{-1}}$, but high densities, so a paved surface stores a lot of energy per square metre. It charges slowly in the morning, keeps warming all afternoon, and stays warm long after sunset. Grass on damp soil behaves quite differently, because a good part of its energy leaves by evaporating water.

The consequence is counter-intuitive and worth telling students: **the urban heat island — the difference in air temperature between a town and the villages around it — is usually largest a few hours after sunset**, not at noon. At noon the town and the countryside are both in full sun; at night the countryside cools quickly, while the town is still giving back the energy stored during the day in asphalt, concrete and brick. A walk at two in the afternoon therefore measures a real effect, but not the maximum one. This is also why the time must be noted with every reading.

5. Lesson flow

The app's wizard has **eleven steps**: (1) the introduction to heat islands and the protocol; (2) a **written prediction**; (3) the **calibration** with the two boxes side by side; (4) spot 1, grass in shade; (5) a second prediction, about the asphalt; (6) spot 2, asphalt in sun; (7) spot 3, indoors versus outdoors; (8) **closing the loop**, back on the grass; (9) the analysis, with the table and its **Corrected (°C)** column; (10) **How sure are we?**; (11) the conclusion about solutions, which quotes the prediction back. Below is what the teacher does around them. Times are for the full two-lesson route; for the short 25-minute route, keep items 1, 4, 5, 6, 7, 9, 10 and 12 below (the guide's numbered items, not the assistant's steps), cut to the times in the time-plan table: the prediction and the vote, without the discussion of reasons (3 min); the protocol read out and posted, not negotiated (2); the calibration with a single box (3); the three spots and the return to the grass (4, 4, 3, 3); and from item 12 only the analysis table, the **How sure are we?** step and the closing sentence (3), with no heat map and no proposal.

1. **The written prediction, before any equipment (10 min).** Hand out the sheets and ask for three written commitments, each with a reason: *which spot in our schoolyard will be warmest, and which coolest?*, *how big will the difference be, in degrees?* and *will indoors be warmer or colder than outdoors?* Take the class vote for the warmest spot and write it on the board.

The numeric predictions are almost always far too large — students frequently say ten or twenty degrees, because they are thinking of the *surface* — and returning to them in the analysis lesson is one of the most productive moments in the whole activity. Do not correct them now. Use the extra minutes to ask for reasons: why does each student think their chosen spot is warmest?

2. **The mechanism (10 min).** Go through the app's first step and turn the four causes it lists into an energy argument: dark surfaces absorb more, buildings block air mixing, human activity adds heat, and missing plants mean missing evaporation. Ask which of the four they think dominates in a schoolyard. Say too what they will *not* measure: the town's heat island proper, which shows when comparing the town with the villages around it, at night.

Technical note — the sensor and the buttons, before the first reading

The sensor. A BME280, which gives temperature, relative humidity and pressure from a single chip. Nothing else connects. At every measurement step — the calibration, the three locations and the return to the grass — the **Temperature** and **Humidity** cards appear, with the **Capture** button and, below them, the chart of **Temperature** and **Relative humidity**, its time in minutes and seconds starting at 0:00 on every step. Above the chart sit the **Smooth**, **Pause** and **Clear** buttons. **Capture** keeps the mean of the last five seconds; until you press it, a reminder appears under the text (you can go on without it, but the row is then missing from the analysis table).

How long to wait. A full two minutes at each spot, plus about five minutes of acclimatisation at the first exit from the building. A box brought from a 22 °C classroom into 12 °C air needs whole minutes to catch up; a reading taken at twenty seconds is the reading of the *previous* spot, and a walk done that way produces a table in which every location resembles the one before it.

How to start it. From the **Activities** page, with the **Start** button on the **Urban Heat Islands** card. The name of the box being followed appears in the activity's header; if several are connected, a small button in the header lets you change it. Name the boxes in **Settings** (for example "Walker" and "Reference") so you do not mix them up.

The buttons you will use. The activity advances with the green button, bottom right, and **its label changes at every step**, because it repeats what you have just finished: **I understand the phenomenon!**, **My prediction is written** (at both prediction steps), **Boxes compared**, **Noted → shade on grass**, **Noted → sun on asphalt**, **Noted → indoors vs. outdoors**, **Loop closed**, **Data analysed**, **Understood** (at the **How sure are we?** step), and at the last step, **Finish**. The **Back** button, on the left, returns one step and keeps what was written at each step separately.

The observation journal. Below the text sits the **observation journal**, a writing box that exists at the measurement steps, at the analysis and at the conclusion; at the two prediction steps, the answer goes into the prediction's own box. What is written there — predictions and captured values included — is saved to disk, in the `logs` folder, and as a **Journal** sheet in the activity's Excel file. It is the right place for the time of the reading and a description of the location.

Exporting the data. At the steps with a chart, two buttons appear: **PNG** saves the chart as an image, and **Excel** saves that step's raw, unfiltered readings into an `.xlsx` file, with an `elapsed_s` column. **There is no CSV button in the activity.** The whole recording is taken from the app's **Data & Export** page: the **Excel** button on the box's row opens the **Export data** dialog, where CSV can also be chosen. Save files using the pattern `class_date_activity` — for example `6B_2026-09-14_insule_termice`. Record on the sheet the time of each reading, the surface, sun or shade, the height used and the weather.

3. **Designing the route (15 min).** Each group marks six spots on the printed plan and justifies

each choice in one sentence. Ask for pairs that contrast in a single thing: grass in sun versus grass in shade shows what shade does; asphalt in sun versus grass in sun shows what the surface does. A route of six spots chosen at random answers no question at all.

4. **The shared protocol (10 min).** Negotiate it with the class and write it on the board, where it stays for the whole activity. There are **five rules**, and none is optional:

   (a) **The same height**, 1.2 m, measured with the tape, at every location.

   (b) **The same shading**: the white card screen above the sensor *at every location*, including the ones already shaded.

   (c) **A full two minutes** of timed waiting before reading.

   (d) **A two-metre exclusion circle**: nobody stands closer to the sensor while it settles.

   (e) **The time noted** with every reading, to the minute.

   Say plainly why: outdoors, nothing can be held constant, so the only thing we can keep constant is our own procedure. If the protocol differs from one location to another, no comparison is legitimate any more.

   Close the lesson with a five-minute **"cold" rehearsal**, right by the classroom door: a single reading taken by the protocol, with the stopwatch running and the exclusion circle respected. A protocol rehearsed once indoors holds up in the field; one merely read out does not.

5. **Calibration and the reference station (8 min).** First, **the two boxes side by side**, in the shade, at the same height, for five minutes. This is the wizard's calibration step: if the second box is connected to the same laptop, the app reads it too and writes a line above the chart such as *"Walker: 21.4 °C · Reference: 21.6 °C → offset −0.2 °C"*; with only one box connected, it asks you to compare it by hand with the reference box or with a thermometer. Press **Capture** when both numbers are steady, then **Boxes compared**. The offset is the difference between the two instruments, subtracted from any comparison between them, and its size shows how small a difference you can believe; the app quotes it in the conclusion. Then start the fixed box in a shaded, ventilated spot, away from walls, and let it record. Explain what it is for: *the whole* location warms up over the course of the morning, and without this record the class cannot tell a warm location from a reading taken late.

6. **Spot 1 — grass in shade (4 min).** This is the app's "grass in shade" step. It is the reference location and must be both first and last in the loop. Give it the full two minutes; students will want to leave after twenty seconds, and the number is still falling. Record the temperature, humidity, time and a one-line description of the surface.

7. **Spot 2 — asphalt in sun (4 min).** First the wizard asks for the second prediction: *how many degrees warmer will the air at* 1.2 *m be over sunny asphalt?* Then comes the "asphalt in sun" step. The sensor sits at the agreed height, shaded by the white card, and **never placed on the hot asphalt**. If you have an infrared thermometer, take the surface temperature here too. Read the technical note below *before* this step.

Technical note — right at the asphalt step: height and the two temperatures

The same height as on the grass: 1.2 m. The step's text says so explicitly, and adds never to lay the box on the hot asphalt. (Versions of the app before 5.3 asked for 10 cm above the ground here; if the text on screen speaks of ten centimetres, update the app.) Ten centimetres above the asphalt is a *different* body of air from the one at 1.2 m — the vertical gradient above a hot surface is 1 to 3 °C between the ground and one and a half metres —, and a group that measured grass at 1.2 m and asphalt at 10 cm would report a difference much of which is only a difference in height.

The 10 cm reading, as an extension. It can be taken *in addition*, as your own observation, and written on its own row of the sheet, below the table — it shows the very existence of the gradient, provided it is never mixed into the table with the others and never captured in the app in place of the one at 1.2 m.
Air, not surface. The **Temperature** card shows the temperature of the *air around the box*. The step's text says asphalt can reach 50 °C in summer while the air above it is only a few degrees warmer than over grass. Students will expect ten degrees and find three. **Say this before they read, not after**, or they will mistakenly mark a correct result as wrong. With an infrared thermometer, note both numbers side by side; the contrast makes the lesson.
Two things changed at once. Grass in shade against asphalt in sun differs in both surface and sun — and the app says so at this very step. Its difference says that something matters, not what. That is exactly why the full route also has grass in sun and asphalt in shade: each changes a single thing relative to the reference.
Humidity drops above asphalt, and that is not a fault. The air there is warmer, so it can hold more water, so the displayed percentage falls — the air has not dried out, it has become thirstier. Above grass, where water really is evaporating, humidity is 5–15 percentage points higher. This is the witness of the mechanism.

8. **The remaining locations (12 min — three stops).** Continue the loop: grass in sun, asphalt in shade, beside the sunlit wall — each changing a single thing relative to an earlier stop. The group moves quickly between locations and stands still during measurement. **A group of students huddled around the sensor is itself a source of heat** — which is why the two-metre exclusion circle is a rule, not a suggestion.

9. **Indoors versus outdoors (4 min).** This is the app's "indoors versus outdoors" step, with the same chart and the same capture button as outdoors. Note the classroom or corridor and write explicitly **whether heating, air conditioning or open windows were on**. Without this note, the reading can never be interpreted again.

10. **Closing the loop (5 min).** This is the wizard's "site 1 again" step, finished with **Loop closed**. Return to spot 1 and measure again, exactly as the first time. **The difference between the first and last reading at the same spot is the warming that happened during the walk, measured instead of assumed.** Its share, in proportion to the time each location was measured, is subtracted from every location in between — the app does the sum itself, in the **Corrected (°C)** column. This step is what raises the activity from an outing to a method, and it is exactly the idea the multi-box app solves differently, by measuring everything at the same time.

Technical note — at the analysis step, where the on-screen table appears

The table fills itself in. At the analysis step, a table replaces the chart, with **four rows** — **Shade / grass**, **Sun / asphalt**, **Indoors** and **Shade / grass (again)** — and the columns **Location**, **Temp. (°C)**, **Humidity (%)** and **Corrected (°C)**, filled from the values captured at the measurement steps. **Corrected** is the reading from which the app has removed the warming during the walk, in proportion to the time at which each value was captured: the starting row stays unchanged, and the return row comes down exactly to the starting value. Below the table, two lines give the differences: grass in shade → asphalt in sun (and its corrected version), grass in shade → indoors, and the drift during the walk. A row with no captured value keeps a dash.
The table is a discussion aid, not the record. Cells can be edited with a double click, but the changes are not kept: **the record is the paper sheet**. The captured values go into the activity's journal anyway.

There is no chart at this step, so no PNG or Excel buttons either. **Export from the measurement steps**, where the chart really exists, or take the whole recording from the **Data & Export** page.
The activity tracks a single box, the one whose name appears in the header; the second box is read only at the calibration step. The reference station is therefore read separately — from the **Live Data** page or from its own laptop — or, more simply, noted by hand at departure and at return. Simultaneous comparison of several boxes is exactly what the *Heat Islands with Several Boxes* app does.

11. **The heat map (15 min, lesson 3).** It starts from the wizard's analysis step, with the table and its **Corrected** column. Each group writes the *corrected* temperatures on the schoolyard plan, then the class combines them: each spot is coloured from blue to red, or simple contour lines are drawn. The warmest and coolest spots are identified, and the two differences the app asks for are calculated — grass in shade to asphalt in sun, and indoors to outdoors.

12. **How sure we are, and the proposal (12 min, lesson 3).** The wizard's **How sure are we?** step asks the class to compare the **drift** (spot 1 at the end minus spot 1 at the start) and the **offset** of the two boxes with the difference between locations: if either is larger, the conclusion is "not detectable with our setup", not "there is no difference" (see the investigation box). Close with the conclusion step, which quotes the group's prediction beside their numbers — the drift and the offset included — and whose green button reads **Finish**. Each group must propose *one* concrete change for the schoolyard — a tree here, a lighter-coloured surface there, a pergola over the play area — and justify it with a number from their own data. If the proposals are good, send them to the school administrator or the town hall: they are real evidence about a real place, and treating them as such completely changes how seriously students measure.

The investigation, in one table

The question. How much warmer is the air above sunny asphalt than the air above shaded grass, in our schoolyard?
The hypothesis. Written on the sheet **before** any measurement: the warmest spot, the coolest spot, and **a number with a unit** for the difference between them. Not changed afterwards.
What I change. One thing only between two compared stops: **the surface** (grass or asphalt) *or* **the sun exposure**, not both at once. The grass-in-shade–asphalt-in-sun pair changes both and says only that something matters.
What I measure. The temperature of the **air**, in °C, to one decimal place, and the relative humidity, in %, at each location. **Not the surface temperature:** these are two different quantities, and the surface one is measured only with an infrared thermometer and noted on its own row.
What I keep the same. The same box from start to finish, compared at departure with the reference station, five minutes side by side, the same 1.2 m height measured with the tape, the white screen at every location including the shaded ones, the same two minutes of waiting, the two-metre exclusion circle, the same person reading the screen, and the time noted with every reading.
How many repeats. Spot 1 is measured **twice**: at the start and at the end of the walk. The locations in between need only one reading after settling: they give the ranking, not the scatter.
What makes a result valid — and what makes it an artefact. Compare **the drift** — the difference between the two readings at spot 1 — and **the offset** from the calibration with the difference between locations, exactly like the app's **How sure are we?** step. **If the drift measured at the same spot or the offset between the boxes is bigger**

than the difference between two locations, the correct conclusion is that the difference is not detectable with our setup — not that it does not exist. These are two different statements, and the distinction between them is exactly what is being taught here. The remedy is not to look for better numbers in the table, but to increase the signal (a different hour, a clear and calm day, locations that contrast more) or to reduce the noise (a shorter walk, a reference station, a second walk in reverse order, averaged with the first).

The link to the app. In *Heat Islands with Several Boxes*, the same rule applies with a number measured directly: the boxes sit side by side for five minutes before setting off, and their scatter is the group's *detection limit*. What you do here with the closed loop, there you do with co-location; the idea is identical, and that is exactly why this exercise is done first.

6. Analysis after the activity

6.1 What the data should look like

The values below are for a walk on a clear, calm day, in the early afternoon, with the sensor correctly shaded, at 1.2 m. All temperatures are **air** temperatures.

Location	Versus reference	Interpretation
Grass in shade (reference)	starting value	Evaporation and shade together; usually the coolest spot. Highest humidity, 5–15 percentage points above asphalt.
Asphalt in shade	0.3 to 1.0 °C warmer	Shade helps, but heat stored in the slab still warms the air.
Grass in sun	0.5 to 1.5 °C warmer	Sun, but no thermal mass and evaporation still going. Sun against shade, screened: 0.5–2 °C.
Asphalt in sun	1 to 3 °C warmer	The main result — but it changes two things at once. Larger on warm, windless days; almost nothing in wind.
Beside a sunlit wall	1 to 4 °C warmer	Reflected and re-emitted radiation, plus sheltered air. Often the warmest spot in the whole schoolyard — and the spot where an unshaded sensor errs most.
The same box, **unscreened**, in the sun	3 to 8 °C above the air	**Radiation error**: the warmed casing, not the air. Not a heat island.
The two boxes side by side, at calibration	usually under 0.5 °C	The offset between instruments; it is subtracted, and it is the smallest difference you can believe.
Indoors, unheated, summer	1 to 4 °C cooler	The building's mass keeps the indoor air behind the outdoor one.
Indoors, with heating or air conditioning	anything	Dominated by the building's systems, not the climate. Note which was running.
Asphalt *surface* temperature	25 to 35 °C above the air	Infrared thermometer only. Compare how much it exceeds the difference measured in air.
Reference-station drift	0.3 to 1.5 °C	Subtracted before comparing locations. On a sunny morning it can exceed the differences between locations on its own.
Windy or overcast day, any pair	usually under 1 °C	Air mixing erases the microclimate. This is a result, not a failure.

The absolute values depend entirely on the day. What any group should reproduce are the *ranking* of locations and the *sign* of each difference. If a group finds asphalt cooler than grass, the first thing to check is whether the readings were taken in a different order, at a different time — exactly what the reference station is for.

6.2 From the table to the map, step by step

1. Record in the table, for each stop, the location number, the time, the temperature to one decimal place and the humidity.
2. Compute the **drift**: the second reading at spot 1 minus the first. Worked example: $23.2 - 22.4 = \mathbf{0.8}\,^{\circ}\,\mathbf{C}$ over the whole walk.
3. Distribute the drift in proportion to the time of the stop — exactly as the app does in the **Corrected** column. A stop made t minutes after the first reading, on a walk of T minutes, gets the correction $0.8 \times t/T$. Worked example, a 30-minute walk, the fourth stop at minute 15: $0.8 \times 15/30 = \mathbf{0.40}\,^{\circ}\mathbf{C}$.

4. Write the corrected temperature. Worked example, again for the fourth stop — asphalt in sun, read as 26.1 °C: $26.1 - 0.40 = \mathbf{25.7\,°C}$.
5. Compute the difference from the reference: $25.7 - 22.4 = \mathbf{3.3\,°C}$. Compare it with the drift of 0.8 °C: the difference is over four times bigger than the drift, so it is a result, not an artefact. Had it come out smaller than the drift, the correct wording would have been *"the difference is not detectable with our setup"*.
6. Transfer the corrected values onto the schoolyard plan and colour it: blue for the coolest, red for the warmest. This is the heat map.
7. **Extension:** walk the same route a second time, in reverse order, and average the two readings at each location. Reverse order cancels a good part of the drift with no calculation at all, and it is exactly the technique used in professional thermal mapping campaigns.

6.3 Questions to ask after the activity

1. You measured the air above the asphalt as about 3 °C warmer than the air above the grass, but the asphalt *surface* was 30 °C hotter than the grass surface. Why is the difference in the air so much smaller?
2. The two readings from the same shaded spot, taken half an hour apart, came out different. What does that tell you about how the readings taken in between must be treated?
3. The grass was both cooler and more humid than the asphalt. Explain in one sentence how a single physical process links these two observations.
4. Two groups walked the same route on the same day, one at 9 o'clock and the other at 14, and got different differences. Give two reasons that do not assume either group made a mistake.
5. The class's prediction was ten degrees, and the result was three. What did your classmates actually predict when they said ten?
6. The asphalt in sun came out warmer than the grass in shade. Can you say the asphalt is the cause? Which other stop on the route helps you answer?
7. A large town's heat island — the difference between the town and the villages around it — is strongest at night. Why, if the sun has set?
8. The school has money for a single change to the schoolyard. Using your own data, argue for an intervention and say what measurement you would take afterwards to check whether it worked.

6.4 Ideas for assessment along the way

- **Low stakes:** the student completes the printed schoolyard plan — each location labelled with its temperature, the warmest and coolest circled, and a single sentence explaining why the warmest is the warmest. No calculation is required.
- **Standard:** a field report that includes the written prediction, the agreed protocol and why each part of it mattered, the table of locations with times, the drift correction applied, the two differences the app asks for, one named source of error with its estimated size, and a conclusion.
- **Extension:** two walks at different times, plotted on the same axes, with a written explanation of how the difference grows over the course of the day; or the use of an infrared thermometer with an explicit comparison between air and surface; or a formal proposal for the schoolyard, with an estimate of the expected cooling, supported by published albedo values.

Common pitfalls and misconceptions

Confusing surface temperature with air temperature. The central trap. Students read that asphalt reaches 60 °C and predict the air will too. Separating the two quantities is a main learning objective, not a detail — and if it is not said beforehand, the class will mark a correct result as wrong.

Leaving the sensor in direct sun. The dominant practical error. An unshaded box, in the sun, reads its own casing and can show 3 to 8 °C too much, manufacturing a heat island that does not exist. Shade *every* location, including the already-shaded ones, so the treatment is identical.

Two things changed at once. Grass in shade against asphalt in sun differs in both surface and sun. The conclusion "asphalt warms the air" also needs the asphalt-in-shade–grass-in-shade pair.

"Our schoolyard is an urban heat island." The schoolyard has microclimates. The urban heat island is the difference in air temperature between town and country, largest at night; the schoolyard shows its mechanisms, not its size.

Different heights at different locations. Above a hot surface, the vertical gradient is 1–3 °C between the ground and 1.5 m. One agreed height, measured with the tape, at every location — 1.2 m, as the app also asks at every step. A reading at 10 cm above the asphalt is taken separately, not in place of the one at 1.2 m.

Waiting too briefly. A box carried from a warm hand into shade needs about two minutes. Readings at twenty seconds record the location the box just came from and produce a table in which every location resembles the one before it.

Holding the box in your hand. Body heat is worth several degrees. The box sits on a stand, not in a palm, and whoever carries it does not read it at the same time.

Crowding around the sensor. Ten students at arm's length are a serious source of heat. The two-metre exclusion circle is set before going out, and it is enforced.

Ignoring the time. On a sunny morning, the whole location can warm up by more than a degree in twenty minutes. Without the reference station and without the closed loop, students will blame that warming on whichever location they happened to reach last.

Measuring near an air-conditioning vent, a kitchen extractor or a car with its engine running. Gives dramatic readings that say nothing about surfaces. Locations are checked the day before.

"The effect is not real, we only got 0.4 °C." On a windy or overcast day this is the correct answer, and it is a result about air mixing and radiation, not a failure. Prepare the class beforehand for this possibility, so a small number is interesting, not demoralising.

Assuming the heat island is largest at noon. It usually peaks a few hours after sunset, because of heat stored in dense materials. A school walk cannot see this directly, but students should know it.

Safety and classroom management

The activity takes a class outdoors, on foot, in exactly the conditions that give the best data — that is, on warm, sunny weather. Both the movement and the weather call for supervision.

Sun and heat. The ideal day for measurement is also a day with a risk of heat exhaustion. Require hats and water, apply sunscreen before going out, **keep total time outdoors under about forty minutes**, use shaded spots as natural breaks, and watch for flushed faces, headaches and dizziness. Students with asthma, heat sensitivity or other relevant conditions are identified beforehand, and their medication goes with the group. In a heatwave, move the walk to the morning or postpone it.

Traffic and crossing. Car parks, delivery ramps, service yards and the school gate are where accidents happen. Reflective vests for any route that touches them, an adult positioned

between the group and any vehicle access, and no measuring spot in a manoeuvring area. Say it plainly: **someone looking at a screen does not see a reversing van**. If the route leaves the school grounds, the school's full off-site procedures apply: parental consent, a named responsible teacher, a first-aid kit, a charged phone, a headcount at every stop and a written risk assessment.

Supervision. For a walk within the closed school grounds, one adult per group of six to eight students is appropriate, with a minimum of two adults in the field, so that one can take a student back into the building without leaving the rest unsupervised. Groups stay within an adult's sight at all times.

Slips and trips. Wet grass, mossy paving, kerbs and steps cause more incidents here than anything else. Walk, do not run, between locations, and whoever is writing stands still. Do not measure during rain or immediately after it, and never in a storm or strong wind — both spoil the data and the safety.

Hot surfaces. Asphalt in full sun can reach temperatures that burn skin. Nobody kneels on it, sits on it or puts a hand on it, and the sensor is held above it, never rested on it — exactly as the app itself says.

Equipment. Laptops and boxes carried outdoors are damaged by falls and unexpected rain. Carry them in a backpack between locations, and agree a quick-retreat plan into the building before going out. **No box is ever left unattended** — not even the reference station: it stays in a visible spot, agreed with the site administrator, and one named student is responsible for it.

Adaptations for pupils with special educational needs

Three locations instead of six, with the plan already numbered. This is the support route and it reaches the same scientific conclusion. The student measures exactly the locations the app proposes — grass in shade, asphalt in sun, indoors — on a printed plan with the locations already numbered and in a table with the names already filled in, so that the task left is: read, record, subtract. Ask only for the **sign and the approximate size** of the difference, and give the drift correction already applied — the **Corrected** column of the app's analysis table gives it ready-made.

The map stays, for everyone. Do not remove the schoolyard plan to simplify the task: for students who need spatial structure, the printed map is exactly the support, not an extra requirement. A table without a map is harder, not easier. For extra support, colour one of the locations yourself *before* copying the sheet: the student continues a pattern rather than inventing one.

Three real roles for students with motor difficulties. The walk does not require everyone to cover the whole route. Roles rotate at every stop, and two of them can be held while standing still:

- **the timekeeper** — starts the stopwatch, announces when the two minutes have passed, and **has the right to reject any reading taken too soon**; this is the role with the greatest scientific weight in the group, because they decide which reading is valid;
- **the data keeper** — stays at a fixed point, next to the reference station, receives the values from each location through classmates, records them in the table with their time, and checks on the spot whether the order of locations and the times agree;
- **the screen reader** — says out loud when the number has stopped and dictates the two values.

A classmate can hold the sensor at the student's direction: "a hand's width higher" and "further from the wall" are experimental decisions too, and whoever makes them is directing the experiment.

Oral answers and drawing instead of writing. The hypothesis, the explanation and

the closing sentence can be given **out loud**, with the teacher or a classmate writing them on the sheet. **The coloured map is a fully accepted answer** in place of the paragraph: a correctly coloured plan, with the warmest and coolest spots marked, shows exactly the same understanding.

Language and cognitive support. Write on the board, at the start, the activity's four words — surface, air, shade, evaporation — and use them unchanged for the whole lesson. Repeat the same sentence every time: *we measure the air above the surface, not the surface.* For students sensitive to crowding and noise, the two-metre exclusion circle is an advantage: give them a fixed waiting spot, in the shade, at every stop.

If things do not go to plan

Rain, wind or overcast — the indoor route. This is not a weak backup, but a whole activity: microclimates exist inside the building too. Four locations, at the same time, with exactly the same protocol: the **sunlit window sill**, the **middle of the room**, the **corridor** and a **north-facing wall** or a room without sun. Wait the two minutes, keep the same height, note the time and note whether the heating was on. Typical differences are 1 to 4 °C, so comfortably measurable, and the mechanisms are the same — received radiation, thermal mass, air that mixes or does not. The loop closes the same way: return to the first spot. If the box also has a light sensor, record how bright each location is too, from the figure on the **Live Data** page: the link between received light and temperature is then visible directly.

You have one box and thirty students. This is actually the normal setup: the activity is built for one box that walks the route. Do the walk once, with the whole class, as a single movement. At each spot, a different student reads the screen out loud and another confirms the number has stopped; everyone records in their own table. You get a single data set, but the whole class analyses it, and the roles rotate around.

You have twenty minutes, not fifty. Keep exactly four moments: the written prediction, one reading on grass in shade, one reading on asphalt in sun, and the closing sentence said out loud. Choose two locations thirty paces apart, right by the door. Drop the route, the reference station and the map — but **do not drop the two minutes of waiting**, because without them nothing is left.

You have no projector. It is not needed: the screen is read outdoors anyway, from close by. Indoors, form a semicircle around the laptop and name a **screen reader** at each location, who says out loud what the number is doing: "still falling", "it has stopped". Describing it out loud is exactly the skill being targeted.

The sensor will not settle. This is normal in the first two minutes, especially at the first exit from the building. If the number is still drifting after five minutes, the cause is almost certainly exposure: direct sun on the casing, the sensor held in a hand, a warm draught from a vent. Put the box on a stand, shade it with the white card and move two metres away. If it still drifts and you came from 22 °C into 5 °C air, simply give it another five minutes: the drop is real, not a fault.

The box will not connect. Do not stop the lesson. Use the "What the data should look like" table in this guide as a printed data set: students write their predictions first, then receive the values row by row, apply the drift correction, compute the two differences and colour the map. The discussion about the air above the surface stays unchanged. Retry the measurement on another day, after checking the cable, the port selected in the app and, if you are on WiFi, the network — outdoors, WiFi is the more fragile of the two anyway.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are root addresses, not deep links, so they will not rot over time.

- **National Meteorological Administration (ANM)** — `https://www.meteoromania.ro` — search for air temperature and ground-observation standards; explains why official thermometers sit in a white, ventilated screen, at a fixed height, and so justifies the protocol students write.
- **US Environmental Protection Agency** — `https://www.epa.gov` — search for urban heat islands; the most complete practical resource on causes, measurement and solutions, and the source of evidence for students' proposals about light-coloured roofing and street trees.
- **NASA Global Climate Change** — `https://climate.nasa.gov` — search for urban heat islands and land surface temperature; satellite thermal images give the large-scale picture of which the class's six locations are a miniature model — and show very clearly that there the subject is *surface*.
- **NOAA** — `https://www.noaa.gov` — search for heat-island mapping campaigns; NOAA runs volunteer campaigns using exactly the route method this activity simplifies.
- **Copernicus** — `https://www.copernicus.eu` — European data on surface temperature and urban climate, letting the class compare its own results with the nearest large city.
- **STEM Learning** — `https://www.stem.org.uk` — search for microclimate and geography fieldwork; classroom-tested sheets and risk-assessment templates for outdoor activities.

8. Assessment

The exit ticket, identical to the one on the student sheet: *one thing that surprised you today, and one thing you would do differently if you repeated the measurement.* It is collected in the last two minutes of the lesson and says more about the class's understanding than any mark.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Explorer level, the three strands read as follows:

- **Measurement practice** — the student follows the protocol without reminders: the same height measured, the white screen at every location, the full two minutes, the exclusion circle, the time noted with the number.
- **Data interpretation** — the student computes the differences with their unit, applies the drift correction, places the values on the map, and describes the ranking of surfaces, not just the largest value.
- **Scientific reasoning** — the student distinguishes air temperature from surface temperature, explains the cooling above grass through evaporation, recognises the radiation error of an unshaded box and a comparison in which two things changed at once, and distinguishes "the difference is not detectable with our setup" from "the difference does not exist".

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. The student sheet

The sheet students fill in is a separate document, named `urban_heat_islands.pdf`, in the folder `docs/en/student_sheets/`. It has two pages — the protocol, the prediction and the walk table on the first; the printed map, the calibration, the drift correction, the two differences and the conclusion on the second — so it prints double-sided, one sheet per student. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — and asks for the prediction **before** any measurement.

Three things worth knowing before printing it. **The five protocol rules are printed at the top of the first page**, where the student has them in view at every stop; they also appear on the

board and in this guide's investigation box, because a protocol heard once is not followed. **The table has its rows already labelled**, in the app's order, with spot 1 marked as the reference, the return row printed as compulsory, not optional, and a **Corrected** column, as on screen; the reading at 10 cm above the asphalt sits *below* the table, on its own row, precisely so it is not mixed with the ones at 1.2 m. The conclusion asks for both the drift and the offset, as at the **How sure are we?** step. **The map is printed, not drawn by the student**: the sheet carries a schematic plan with the six zones, each with its location number and its temperature. The real plan of the schoolyard, if you print one, is glued to the back of the sheet. For students who need spatial structure, the printed map is the main support of the whole activity. The tables are not repeated here precisely so that the two cannot drift apart: there is a single printable version, and it is the sheet.

Once the class has mastered this protocol, move on to the *Heat Islands with Several Boxes* app: the same locations, measured at the same moment with several boxes, and a detection limit that is measured instead of inferred.